

Beyond the Seven-Eighths Barrier: A Sharper Quasi-Riemann Boundary

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Abstract. We refine the zero-free boundary in OpenAI’s September 30 quasi-Riemann paper by perturbing the total length of its compensation primes and adjusting the two column scales. Starting from that paper’s stated unconditional results, we obtain the strict zero-free half-plane $\operatorname{Re} s > 3499999/4000000$ for finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$ and for all Dirichlet L -functions. The improvement over $7/8$ is $1/4000000$. The same finite probe supports both estimates: its rescaling coefficients absorb the additional completed-row energy, while an explicit central margin survives the perturbation. We give the affected counting, contour, and principal-term arguments in full, including a parameter order that keeps the moment mesh independent of the later principal approximation. The foundational reflection, sieve, and moment theorems remain inputs from the original paper; no new short mean-square assumption is imposed.

1 The boundary and the perturbation

Using Argonaut Math, we derive a parameter refinement of OpenAI’s September 30 argument. The result forms part of the program’s broader mathematical research. The underlying analytic machinery comes from that paper, which proves a zero-free half-plane with boundary $7/8$ and constructs a finite arithmetic probe whose two representations connect a sieve estimate to an analytic signal. We show that a small increase in its compensation-prime length can be accommodated throughout this construction and yields a strictly smaller boundary.

The historical question begins with Riemann’s 1859 paper, which connected the distribution of primes to the zeta function through analytic continuation and a functional equation. In discussing its zeros, Riemann proposed the critical-line hypothesis now bearing his name [2, English translation, pp. 2–4]. The present argument concerns a zero-free half-plane for a family of L -functions; it does not reach the critical line.

We work over $F = \mathbb{Q}(\sqrt{-3})$, with ring of integers $\mathcal{O}_F$ and ideal norm $N\mathfrak{a} = |\mathcal{O}_F/\mathfrak{a}|$. A finite-order Hecke character η is extended by zero to ideals not coprime to its modulus. For $\operatorname{Re} s > 1$ its L -function is

$$L_F(s, \eta) = \prod_{\mathfrak{p}} (1 - \eta(\mathfrak{p})(N\mathfrak{p})^{-s})^{-1}.$$

We use its usual meromorphic continuation. Removing the factors at a fixed finite set S of prime ideals gives $L_F^S(s, \eta)$. The corresponding finite factors are nonzero for $\operatorname{Re} s > 0$. As usual, a principal character may have a pole at $s = 1$.

Theorem 1.1. *All finite-order Hecke L-functions over F , and all Dirichlet L-functions, are zero-free in the strict half-plane*

$$\operatorname{Re} s > \frac{3499999}{4000000}.$$

The pole at $s = 1$ for a principal character is allowed. The boundary is $1/4000000$ smaller than $7/8$; no assertion is made about zeros on the boundary line.

This is an unconditional literature deduction: the inputs are results stated and proved unconditionally in [1], not an additional analytic conjecture. Theorem 1.1 of that paper supplies the existing $7/8$ bound. We also use Propositions 2.1, 11.3, and 15.2; Lemmas 10.2, 11.1, 13.1, 14.3, 17.1, 17.6, 18.1, and 19.1; and the coefficient, reflection, and zero-detection results of Sections 4–8. Whenever a numerical range in the original argument is affected, we give the requisite deduction below from its general energy or moment estimates and exact local formulas. We do not reproduce the full foundational reflection, sieve, or moment proofs. Page references to [1] use its printed PDF pagination, and its equation numbers retain their original meaning. The conclusion here does not include a zero-free boundary below $3/4$.

Let Z tend to infinity. The exponent ℓ is the total compensation-prime length, and l_x, l_y give the column scales $X = Z^{l_x}$ and $Y = Z^{l_y}$. Their sum is M , and their difference is b . The exponent h marks the upper central-row scale before a small buffer is added; σ_τ is the proposed zero-free boundary. Increasing ℓ shortens both columns and moves this upper row scale. We make the following fixed choice:

$$\begin{aligned} \tau &= 10^{-6}, & \ell &= \frac{1}{6} + \tau, & M &= \frac{5}{6} - \tau, & b &= \frac{1}{8}, \\ l_x &= \frac{17}{48} - \frac{\tau}{2}, & l_y &= \frac{23}{48} - \frac{\tau}{2}, & h &= \frac{13}{16} + \frac{3\tau}{2}, \\ \sigma_\tau &= \frac{7}{8} - \frac{\tau}{4} = \frac{3499999}{4000000}. \end{aligned} \tag{1}$$

Thus

$$l_y - l_x = b, \quad l_x + l_y = M, \quad M + \ell = 1, \quad h = 1 - l_x + \ell.$$

For the two column scales take $X = Z^{l_x}$ and $Y = Z^{l_y}$. An especially useful feature of this adjustment is that the affine exponent of the analytic signal does not change:

$$C(s) = \frac{l_x}{2} + s - 1 + \frac{h}{6} = s - \frac{11}{16}, \quad C(\sigma_\tau) = \frac{3}{16} - \frac{\tau}{4}. \tag{2}$$

To avoid assuming that a rightmost zero exists, measure the whole family by its supremum

$$\beta_* = \sup \left(\{1/2\} \cup \{\operatorname{Re} \rho : L_F(\rho, \eta) = 0, 1/2 \leq \operatorname{Re} \rho \leq 1, \eta \text{ primitive of finite order}\} \right).$$

The supremum need not be attained. Throughout the proof we suppose, for a contradiction, that $\beta_* > \sigma_\tau$, and put $\Delta_\tau = \beta_* - \sigma_\tau > 0$. The existing result gives

$$\sigma_\tau < \beta_* \leq \frac{7}{8}, \quad \kappa = \frac{3}{4}, \quad \beta_* \leq \frac{1 + \kappa}{2}. \tag{3}$$

This permits the fixed value $\kappa = 3/4$ in Lemma 18.1 of [1], whose uniform range is $\kappa \in [3/4, 1]$. It is the old bound, already available before the perturbation, that justifies this choice.

The argument compares two estimates for the same finite arithmetic probe. The low-side estimate bounds its size using reflected-row energy and an additive Gram bound; rescaling absorbs the cost of the longer compensation primes. The high-side representation separates a principal contribution from the other rows. After normalization, the principal contribution is the signal in (18), whose integrand contains $H_\eta(s)/L_F^S(s, \eta)$. The role of this reciprocal L -function is to connect continuation of the signal to the location of zeros, through Proposition 2.1 of [1].

Sections 4 and 5 bound the nonprincipal contribution: actual witness capacities provide a row count with an explicit margin, and a joint numerator and error-slot estimate preserves it after the full contour has been moved. Section 6 extracts the principal signal and handles the outer rows. Finally, Section 7 fixes the parameters in a noncircular order and supplies the two estimates required by the continuation principle to exclude the gap Δ_τ .

2 One finite probe and its Euler representation

The probe is the finite weighted arithmetic sum $I_\eta(X, Y, Z)$ of (6.1) in [1], with column scales X, Y and large parameter Z . Its arithmetic form is used for the low-side size bound. Its Euler representation is used to isolate the principal signal and estimate the remaining terms. In that representation the index u labels the physical rows, and $u = 1$ is the principal row.

Each compensation-prime slot selects one prime from a prescribed window. The modification below combines terms in which selected primes impose divisibility conditions with terms in which they rescale the columns. These terms must form the same finite sum in both estimates, with their conjugate phases and coprimality restrictions intact. We therefore specify the finite modification first and derive its integral representation afterward.

Fix a quotient T of a ray class group before choosing a target character. For each target η , use fixed arithmetic data S, b_*, ξ allowed by Sections 6 and 13 of [1]; here b_* is a fixed ideal, and S contains the relevant moduli and small primes. Also fix the nonnegative, nonzero smooth annular weights W_0, W_1 and the Fourier normalization of Section 6 of that paper. Denote its finite physical probe (6.1) by $I_\eta(X, Y, Z)$. Inserting $1_{D|cn^3}$ into its completed rows gives $I_{\eta;D}$. The ideals c, n retain the original coprimality conditions and the characters retain their zero extension. Thus these symbols refer to the finite expressions of (6.1), not to expressions defined by the integral below.

Choose a fixed even integer K and split the compensation length equally: $\ell_i = \ell/K$ and $P_i = Z^{\ell_i}$. Choose K pairwise disjoint compact supports in $(1, 2)$, separated by positive gaps, and nonnegative, nonzero smooth weights W_i on them. The prime window is

$$\mathcal{P}_i(Z) = \{\mathfrak{p} \notin S : \mathfrak{p} \in 1_T, W_i(N\mathfrak{p}/P_i) \neq 0\}.$$

The condition $\mathfrak{p} \in 1_T$ means that its ray class is the identity. For $J \subseteq \{1, \dots, K\}$ write $p_J = \prod_{i \in J} P_i$, with the empty product equal to the unit ideal. The modified probe is the same finite expression as

(12.5) of [1], at the new scales:

$$I_{\eta, \text{mod}}(Z) = \sum_{(\mathfrak{p}_i) \in \prod_i \mathcal{P}_i(Z)} \prod_i W_i(N\mathfrak{p}_i/P_i) \times \sum_{J \subseteq \{1, \dots, K\}} (-1)^{|J|} (Np_J)^{-3/2} \overline{\eta(p_{J^c})} \times I_{\eta; p_{J^c}}(X/Np_J, Y/Np_J, ZNp_{J^c}). \quad (4)$$

Each window retains its original scale P_i . The conjugate in (4) pairs the marked contribution with the rescaled contribution in the local compensation. Identity ray class makes the characters in $\widehat{T}$ equal to 1 at that prime; it does not make $\eta(\mathfrak{p})$ equal to 1. The target may have any unit phase there.

For the Euler representation use the local data $P_{\mathfrak{p}}, P_{\mathfrak{p}}^*, H_{\mathfrak{p}}, V, W, D$ of (7.11)–(7.12) in [1], and denote the complex variable called x in its Section 16 by s . At $Q = N\mathfrak{p}$ the complete local replacement is

$$G_{\mathfrak{p}} = \frac{(\overline{\eta(\mathfrak{p})}Q^s - Q^{-w})(1-V)(1-W)P_{\mathfrak{p}}^* - Q^{-w}(1-VW)}{1-D}. \quad (5)$$

Equation (16.4) of that paper reads

$$\overline{\eta(\mathfrak{p})}Q^{s+z-1}P_{\mathfrak{p}}^* - Q^{z-w-1}P_{\mathfrak{p}} = \frac{1-D}{(1-V)(1-W)}Q^{z-1}G_{\mathfrak{p}}.$$

Consequently the full correction is

$$H_{\eta, u, Z}(s, w, z) = \sum_{(\mathfrak{p}_i) \in \prod_i \mathcal{P}_i(Z)} \prod_i [W_i(N\mathfrak{p}_i/P_i)(N\mathfrak{p}_i)^{z-1}G_{\mathfrak{p}_i}] \times \prod_{\substack{\mathfrak{p} \notin S \\ \mathfrak{p} \in \{\mathfrak{p}_1, \dots, \mathfrak{p}_K\}}} H_{\mathfrak{p}}. \quad (6)$$

Every local factor uses the current physical row u . Formula (5) remains defined at zeros of $P_{\mathfrak{p}}$ or $H_{\mathfrak{p}}$, and is the formula used to continue the complete correction. Disjoint supports ensure that each selected prime is replaced exactly once, even though the nominal scales P_i are equal.

The smooth factors supply decay in the integral variables. Let $\widetilde{W}_0(|y|^2)$ be the plane Fourier transform of $W_0(|y|^2)$ for the self-dual measure of [1], and set

$$\mathcal{M}(z) = \int_0^\infty \widetilde{W}_0(r)r^{z-1} dr, \quad \widehat{W}_1(w) = \int_0^\infty W_1(r)r^{w-1} dr.$$

The resulting weight is

$$\mathcal{W}(X, Y, Z; s, w, z) = X^{1/2-z} Z^{s+z-1} Y^{w-1} e^{(s+z-1)^2} \mathcal{M}(z) \widehat{W}_1(w).$$

Equations (7.5), (7.13), and the finite replacements give the exact identity

$$I_{\eta, \text{mod}}(Z) = \frac{1}{(2\pi i)^3} \int_{\text{Re } s=3} \int_{\text{Re } w=3} \int_{\text{Re } z=2} \mathcal{W}(X, Y, Z; s, w, z) \times \sum_u^{(6)} (Nu)^{-z} \overline{\xi(u)} \frac{\zeta_F^S(6z) L_F^S(w, \chi_\bullet(u))}{L_F^S(s, \eta \chi_\bullet(u))} H_{\eta, u, Z}(s, w, z) dz dw ds. \quad (7)$$

Here (6) preserves the original physical restrictions of (7.5): nonzero sixth-power-free elements, the chosen unit sector, and $(u, S) = 1$. The ideal character $\chi_\bullet(u)$ is the one in Lemma 4.1 of [1]. The initial lines are absolutely convergent, since the identity comes from finitely many absolutely convergent expansions of the original probe. For each fixed Z , the slots, prime tuples, and rescaling subsets are finite. The completed sum over u , the Euler products, and the three Mellin integrals are infinite. Their absolute convergence on these initial lines is the interchange justification in (7.5) of [1]; multiplying by the finitely many fixed-tuple replacements preserves it. Subsequent contour moves require the full-row majorants proved in Sections 5 and 6. No local identity uses $\ell = 1/6$. Thus (7) is still an exact high-side representation of the finite probe (4) at the perturbed geometry.

3 The cost of compensation on the low side

The extra prime length changes the completed rows. The useful estimate must therefore come from the general reflection energy, rather than from a numerical endpoint specialized to the old lengths. Once that energy is known, the additive Gram estimate and the coefficient in (4) show that the cost is absorbed.

Fix a rescaling subset J and write

$$d_J = \sum_{i \in J} \ell_i, \quad \ell' = \ell - d_J, \quad M' = M - 2d_J,$$

$$X' \asymp Z^{\ell_x - d_J}, \quad Y' \asymp Z^{\ell_y - d_J}.$$

Then $0 \leq d_J \leq \ell$ and $M' + \ell' - 1 = -3d_J$. For a real number t , write $t_+ = \max\{t, 0\}$.

Lemma 3.1. *The completed-row factor $B_{m,\sigma}^J$ in (15.1) of [1], where $\sigma \in T$ is the finite ray class label, satisfies*

$$\sum_{0 < Nm \leq Q} |B_{m,\sigma}^J(Z)|^2 \ll Z^{M' + (5\ell - 1 + d_J)_+ / 4 + \varepsilon}, \quad Q \asymp Z^{M'}.$$

The constant may depend on the fixed windows and arithmetic data, and is uniform in the moving rescaling prime tuple.

Proof. Apply the general reflected-energy estimate of Lemma 14.3 in [1]. Its decomposition separates the actual row into a powerful part, a fixed-support part, and a remaining square-free part. Let O be the powerful-part length and H the remaining-row length, and put $\Delta_H = M' - O - H \geq -O(1/\log Z)$. The Gaussian subdivisions, common-profile separation, and tail truncation on pp. 102–103 of that paper leave the length identity unchanged. Denote their arbitrarily small positive loss by ε_0 .

In the notation of Lemma 14.3 we have $2A_0 \leq O + O(1/\log Z)$, $N_0 \leq A_0$, and $z_a \leq \ell'$, with

$$T_{d_J} = 2H + 2A_0 + 2z_a - 1 - \ell' - \theta_N - N_0 - 3B_0.$$

The identity $M' + \ell' - 1 = -3d_J$ gives $T_{d_J} \leq H - 3d_J + O(\varepsilon_0)$. The retained dual scales satisfy $\nu + 3\ell_b + e_\lambda \leq T_{d_J} + \varepsilon_0$, so

$$\max(H, \nu + \ell_b) = H + O(\varepsilon_0).$$

The local parameter called u in that energy lemma is $\min\{v, z_a, (v + z_a)/3\}$; it is independent of the physical-row notation in (7). If this minimum is z_a , the energy is at most $M' - \Delta_H + O(\varepsilon_0)$. Otherwise $u \geq v/2 - O(\varepsilon_0)$ and

$$u + \ell_b + \frac{2e_\lambda}{3} + \frac{(T_{d_J} - v - 3\ell_b - e_\lambda)_+}{2} \geq \frac{T_{d_J}}{4} - O(\varepsilon_0).$$

Substitute this in (14.14), using the nonnegative second numerator in (15.3). The other energy bound is

$$M' + \frac{1 + 3\ell' - 2M' - 2\Delta_H}{4} + O(\varepsilon_0).$$

Both bounds are nonincreasing in Δ_H . We may therefore put it equal to zero and absorb its actual boundary error. Since $1 + 3\ell' - 2M' = 5\ell - 1 + d_J$, the two branches give the asserted positive-part bound. The common-profile kernel amplitude is extracted before squaring. Powerful-row counting is already included in Lemma 14.3, and its kernel saving is evaluated on the actual dual support. Small losses, finitely many subsets, and logarithmically many actual scales fit the stated ε .

The moved endpoint deserves separate attention. When $d_J = \ell$, $\ell' = z_a = 0$ and

$$T_{d_J} \leq 1 - 6\ell + O(\varepsilon_0) = -6\tau + O(\varepsilon_0).$$

A nonempty retained dual block has length exponent at least $-O(1/\log Z)$. First choose the truncation loss sufficiently smaller than τ , and then let Z be sufficiently large. The retained block is empty. The discarded part is controlled by the uniform tail estimate in Lemma 14.3, while the first branch also gives the claimed bound. This handles the new zero-length endpoint directly. $\square$

Lemma 3.2. *For every $\varepsilon > 0$,*

$$|I_{\eta, \text{mod}}(Z)| \ll Z^{C(\sigma_\tau) + \varepsilon}.$$

Proof. Set $P_a = Y'^2/Q = (Nb_*)^{-1}Z^b$. The scales for a fixed tuple satisfy $Q, Y' \geq 1$ and $P_a \geq 1$: indeed

$$M - 2\ell = \frac{1}{2} - 3\tau > 0, \quad l_y - \ell = \frac{5}{16} - \frac{3\tau}{2} > 0.$$

The additive Gram bound of Proposition 15.2 in [1] permits these general polynomial scales. Moreover,

$$2b - l_y + d_J \leq -\frac{1}{16} + \frac{3\tau}{2} < 0.$$

Hence its P_a^2/Y' term is absorbed by the $P_a^{1/6}$ term, and the squared Gram norm is at most $Z^\varepsilon (Q/Y') P_a^{1/6}$ times a fixed height polynomial.

Cauchy–Schwarz with Lemma 3.1, including the $Q^{-1/2}$ factor in the original separation formula, gives the fixed-tuple exponent

$$\frac{l_x - d_J}{2} + \frac{b}{12} + \frac{(5\ell - 1 + d_J)_+}{8}.$$

There are at most $Z^{d_J + \varepsilon}$ rescaling tuples, while their coefficient in (4) supplies $Z^{-3d_J/2}$. Thus the total exponent is

$$L - d_J + \frac{(5\ell - 1 + d_J)_+}{8}, \quad L = \frac{l_x}{2} + \frac{b}{12} = C(\sigma_\tau).$$

Because $\ell < 1/5$, the function $-d_J + (5\ell - 1 + d_J)_+/8$ starts at zero and is strictly decreasing; its two slopes are -1 and $-7/8$. Every subset therefore has exponent at most L . Rapid decay of the smooth weights absorbs the fixed height polynomials, and the total small loss can be chosen below the prescribed ε . $\square$

4 Witness capacity and the central margin

The high-side rows are measured on a different scale from the rescaling subsets of the previous section. Write $Nu \asymp U = Z^d$, so that d is a row-norm exponent, and fix $d_{\min} = 1/100$. In a zero-detection box, let R denote the ideal row-count exponent. We will prove a bound $\#\mathcal{B} \ll_{\eta} U^{R+\varepsilon_{\text{cnt}}}$ only after paying the raw moment, witness, rounding, logarithmic, and height costs. The allowance ε_{cnt} is not part of R .

The source's zero-detection argument supplies inverse and ordinary witnesses whose moment estimates give two bounds for this count. We use the compensation primes already present in the probe to strengthen those bounds. The capacity of a moment estimate means the total prime-slot length that its hypotheses permit. The selected slots must fit that capacity, and their phases must agree with the witnesses. We first arrange those phases, then compare the two resulting counts at their crossing point.

As in Section 6.1 of [1], put $\Theta = \langle \eta, \widehat{T} \rangle$. Within each target, this finite character group and every slot coefficient are fixed independently of the moving row. The direction of the witnesses is handled in two steps, following Section 19.2, pp. 182–183. First, if necessary, conjugate the two witnesses and their smooth profiles so that their common character is

$$\psi(n) = \nu(n) \overline{\chi_n(u)}, \quad \nu \in \Theta.$$

Relative to this character the physical prime-slot coefficient is

$$\overline{\nu(\mathfrak{p})} 1_{\mathfrak{p} \in 1_T} = \frac{1}{|T|} \sum_{\mathfrak{g} \in \widehat{T}} (\overline{\nu} \mathfrak{g})(\mathfrak{p}).$$

It is a fixed combination of Θ -characters, and the marked moment permits this negative direction. For the positive-direction ordinary moment in Lemma 18.1, conjugate the entire product of the two ordinary witnesses and all selected physical slots. Its absolute square is unchanged; the common character becomes $\overline{\nu}(n) \chi_n(u)$ and the slot coefficient becomes $\nu(\mathfrak{p}) 1_{\mathfrak{p} \in 1_T}$. This is a conjugation of the whole product, not a change of phase in an individual factor.

Both steps preserve zero extension, original prime support, and independence of the profiles. Since Θ is a group, membership of the induced character in Θ is unchanged. Every retained large row ramifies at a prime outside S , so its induced character is outside Θ , as Lemma 18.1 requires when the slot length is positive. Witness heights and physical heights need not coincide. After fixing the physical height and the selected slots, Lemma 4.5 handles the witness smooth parameters at the cost of a fixed height power, uniform in the moving rows for the fixed arithmetic datum. Indeed the physical normalized slot is

$$P_i^{-1/2} \sum_{\mathfrak{p} \in \mathcal{P}_i(Z)} \overline{\chi_{\mathfrak{p}}(u)} W_i(N\mathfrak{p}/P_i) (N\mathfrak{p}/P_i)^{z-1}.$$

Its annular profile is $W_i(y)y^{z-1}$ and its base- U length is ℓ_i/d , since $U^{\ell_i/d} = P_i$. The character expansions above identify exactly this polynomial with the prime factors in Lemmas 17.1 and 18.1. The disjoint supports are those fixed in (4), before the original zero masks are imposed. Equal nominal lengths do not merge those supports. Thus the cited moment inputs apply to the actual windows and profiles of this probe.

In a nonprincipal dynamic zero-detection box let $a \leq \beta_*$ be its real-part parameter, and put $\delta = 2a - 1$. Away from the bottom box,

$$\frac{1}{50} < \delta \leq \frac{3}{4};$$

the bottom box uses $a = 51/100$. Lemma 19.1 supplies an average prime-amplitude parameter q satisfying $0 \leq q \leq \delta/2$. Here q is a dimensionless exponent, not a prime norm. For a nonbottom box put $\alpha = 5/6$ and $x = q/\delta \in [0, 1/2]$. The crossing of the two witness counts is described by

$$\begin{aligned} D_x &= 3 - \frac{17x}{9}, & P_x &= \left(2 - \frac{8x}{9}\right)(1-x), \\ J &= (\alpha - \delta)D_x + \delta P_x, \\ R_* &= 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2J}. \end{aligned} \tag{8}$$

We have $D_x \geq 37/18$, and J has a positive lower bound on this compact parameter range. The letter J here is a real denominator; it is unrelated to the rescaling subset in (4).

Lemma 4.1. *Put $c_{\text{sup}} = 1/185$. Suppose that the available base- U prime length is at least $7/37 + c_{\text{sup}}$. Fix positive allowances $\varepsilon_m, \varepsilon_w, \nu_0, \omega_{\text{slot}}$ with*

$$\varepsilon_w \leq 10^{-3}, \quad \max\{\varepsilon_w, \nu_0, \omega_{\text{slot}}\} < c_{\text{sup}}/16.$$

Require ω_{slot} to be no larger than the mesh supplied by Lemma 18.1 of [1] for loss ε_m and the bounded length ranges here, and require $\max_i \ell_i/d \leq \omega_{\text{slot}}$. For each target with its fixed arithmetic data, final excluded set, and fixed annular windows, there exist finite $A_{\text{cnt},\eta} \geq 0$, an integer $L \geq 0$, and a constant $C_\eta > 0$ such that the following raw bound holds in every retained nonbottom dynamic and amplitude bin $\mathcal{B}$ under the detector's loss and height conditions:

$$\#\mathcal{B} \leq C_\eta (1 + \log U)^L (1 + H)^{A_{\text{cnt},\eta}} U^{R_* + \varepsilon_m + 4\varepsilon_w + 6\nu_0 + \omega_{\text{slot}}}. \tag{9}$$

Here $H \geq 1$ bounds the retained physical and witness norm-twist parameters, $\kappa = 3/4$, and $1/50 < \delta \leq 3/4$ as above. The constants and orders may depend on the fixed arithmetic datum, windows, slot count, strict moment margins, and prescribed losses, but not on U, H , the moving rows or masks, the retained point, or any later external tail order. They are fixed before a trial exponent in $T_1 = Z^\theta$ is chosen; the sufficiently large threshold may depend on that exponent. Without selecting any slots, $t = 1$ gives the same type of raw bound with R_ replaced by $1 - 2\delta/3$ and error $\varepsilon_m + 4\varepsilon_w$, without a prime-supply requirement.*

Proof. We redo the capacity calculation in the proof of Section 19.2 rather than apply the specialized statement of Proposition 19.2 with its old positive gap. The inverse witness uses Lemma 17.1, and the ordinary witness uses Lemma 18.1; the latter's requirement $\beta_* \leq (1 + \kappa)/2$ holds by (3).

Choose $t \in [1, 3/2]$ as below, before selecting the witnesses. Proposition 8.3 of [1] supplies a common presentation and actual lengths (r, m) . Write M_r for its normalized inverse witness and S_m for its normalized ordinary witness. Its saturation error constants are absolute on these bounded ranges. Reduce its preliminary loss using those constants and then increase the threshold to absorb the $O(1/\log U)$ errors. This gives

$$r + m \geq t - \varepsilon_w, \quad r \leq t + \varepsilon_w, \quad r \geq t - 1/2 - \varepsilon_w, \quad 0 \leq m \leq 1/2 + \varepsilon_w, \\ |M_r|^2 \gg_\eta U^{\delta r - \varepsilon_w}, \quad |S_m|^2 \gg_\eta U^{\delta m - \varepsilon_w}.$$

For example, if C_* bounds all its saturation error constants, take the preliminary loss at most $\varepsilon_w/[2(C_* + 1)]$; C_* precedes the target. The detector's literal height restrictions are verified in Section 7.

Order the slots by decreasing nonnegative amplitude exponent, and select only positive-exponent slots. If the request is z , a fractional selection has weighted amplitude at least qz : if all positive slots are exhausted, their full amplitude is $q\ell/d \geq qz$. Discarding the last fractional slot loses at most $\delta\omega_{\text{slot}}$ in the squared exponent. Before rounding, reduce a positive moment capacity c to $z = c - \nu_0$; the total loss relative to $2qc$ is at most

$$2q\nu_0 + \delta\omega_{\text{slot}} \leq \nu_0 + \omega_{\text{slot}}.$$

The actual whole-slot length is at most z , even when there are too few positive slots. Zero-exponent and error slots are used only through upper bounds. Use the two direction changes above: first put witnesses and physical slots in the common negative direction, then conjugate the complete ordinary product. The respective coefficients are $\bar{\nu}(\mathbf{p})1_{\mathbf{p} \in 1_T}$ and $\nu(\mathbf{p})1_{\mathbf{p} \in 1_T}$, both independent of the moving row. Zero extensions, independent windows, and the full-support conditions are retained.

The ordinary-witness capacity is exactly

$$\frac{1 - 2m}{6\kappa} = \frac{2(1 - 2m)}{9}.$$

At $\kappa = 3/4$ it equals the baseline capacity used in the crossing calculation of Section 19.2. There is no need for the $\Delta/4$ inflation introduced there by $\kappa = 3/4 + 2\Delta$. Here Q denotes the product of the selected normalized prime polynomials, in the common character direction just fixed. The marked moment at effective row width one bounds $\sum_u |M_r Q|^2$ by $U^{1+\varepsilon_m}$ times its fixed height factor when $r + 2z \leq 1 - c_1$ and $2r + 8z \leq 3 - c_2$. The ordinary moment bounds $\sum_u |S_m^2 Q|^2$ by the same power when $2m + 6\kappa z \leq 1$; its divisor is $|S_m|^4$. At zero slot length its assertion holds for all the bounded m here.

If $r \geq 1$, the no-slot sixth-power amplified moment of Lemma 17.6, equation (17.88), gives exponent

$$1 - \alpha + (\alpha - \delta)r + \varepsilon_m + \varepsilon_w \leq L(t) + \varepsilon_m + (1 + \alpha)\varepsilon_w, \quad L(t) = 1 - \delta + (\alpha - \delta)(t - 1).$$

Its form $e(r) = \max\{1, (1 + 5r)/6\}$ is uniform as r approaches one. If $m \geq 1/2$, the ordinary no-slot count has exponent at most $1 - \delta + \varepsilon_m + 2\varepsilon_w$. In the remaining range $r < 1, m < 1/2$, the two ideal counts at their full capacities are

$$A_I(r) = 1 - \delta\{x + (1 - x)r\},$$

$$S_t(r) = 1 - \delta\{4x/9 + (2 - 8x/9)(t - r)\}.$$

Replacing m by $t - r - \varepsilon_w$ costs at most $2\varepsilon_w$ because the ordinary count decreases with m and its derivative has absolute value at most 2. Its witness divisor costs another $2\varepsilon_w$. The crossing occurs at

$$r_*(t) = \frac{(2 - 8x/9)t - 5x/9}{D_x}, \quad \frac{23}{37} \leq r_*(t) \leq 1, \quad \frac{1}{3} \leq t - r_*(t) \leq \frac{1}{2}.$$

Use the ordinary count for $r \leq r_*(t)$ and the inverse count for $r \geq r_*(t)$. For the latter, if $(1-r)/2 > \nu_0$, request $z = (1-r)/2 - \nu_0$. Its whole-slot length z' satisfies

$$1 - r - 2z' \geq 2\nu_0 > 0, \quad 3 - 2r - 8z' = 4(1 - r - 2z') + (2r - 1) \geq \frac{9}{37}.$$

Thus one may fix $c_1 = 2\nu_0$ and $c_2 = 9/37$ before the target. For the ordinary request $z = 2(1 - 2m)/9 - \nu_0 > 0$, one has $1 - 2m - 6\kappa z' \geq (9/2)\nu_0$. The unreduced inverse capacity is at most $7/37$, and the ordinary capacity in its chosen branch is at most $2/27 + 4\varepsilon_w/9$. Both are strictly below the available supply with the fixed slack specified in the statement. Decrement, witness error, and slot mesh are each smaller than $c_{\text{sup}}/16$; no limiting equality is used.

The common short-witness bound is

$$R_{\text{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t).$$

Choose

$$t = 1 + \frac{\delta P_x}{2J} \in [1, 3/2],$$

which makes $R_{\text{short}}(t) = L(t)$ and yields exactly R_* in (8). If the inverse capacity is at most ν_0 , the no-slot ideal exponent is at most $1 - \delta + 2\delta\nu_0$; if the ordinary capacity is at most ν_0 , it is at most $1 - \delta + (9/2)\delta\nu_0$. Since $R_* \geq 1 - \delta$, these cases fit the common error $\varepsilon_m + 4\varepsilon_w + 6\nu_0 + \omega_{\text{slot}}$. For $q = 0$ the slot spike is 1 and the same argument uses no amplitude division. Empty bins contribute zero; a vanishing slot cannot belong to a positive-amplitude class.

For completeness, without slots take $t = 1$. For $r \leq 1$, the two ideal exponents are $1 - \delta r$ and $1 - 2\delta(1 - r)$, whose smaller value is at most $1 - 2\delta/3$, with crossing $r = 2/3$. For $r > 1$, the amplified estimate is at most $1 - \delta$ up to the displayed witness errors. This proves the asserted no-slot form.

We now keep the height and multiplicity factors that these moments actually supply. Lemmas 17.1, 17.6, and 18.1 give finite seminorm and height orders for the fixed datum, uniformly in their bounded real ranges, masks, and moving supports. Their application includes the finite derivative profiles required by the parameter Sobolev inequality of Lemma 4.5. A logarithmic scale derivative inserts $yW'(y)$; a normalized twist derivative inserts $\log y$ on the same annulus. Sum over rows before the nonnegative Sobolev integrals. If a particular fixed-parameter moment has height degree J_j and h_j height coordinates, covering their box costs at most $(1 + H)^{J_j + h_j}$. Take $A_{\text{cnt}, \eta} = \max_j (J_j + h_j)$ over these actual calls, their derivative profiles, and the no-slot calls. The finite presentation labels and slot subsets enter C_η ; witness dyads and logarithmic scale coverings contribute $(1 + \log U)^L$. This finite maximum is obtained from the stated moment theorems, not assumed as an extra analytic input. It need not be uniform in η . All its orders and constants precede the trial height exponent. Auxiliary external tails in the detector are made smaller than a fixed fraction of its spike after that exponent is chosen; their constants only enlarge the lower threshold. They do not change these internal moment orders. This proves (9) with the asserted dependencies. $\square$

To obtain a pure- U bound, reserve positive $\varepsilon_{\log}$ and $\varepsilon_{\text{cnt},\text{ht}}$ within the count allowance, with

$$\varepsilon_m + 4\varepsilon_w + 6\nu_0 + \omega_{\text{slot}} + \varepsilon_{\log} + \varepsilon_{\text{cnt},\text{ht}} < \varepsilon_{\text{cnt}}. \quad (10)$$

The logarithm in (9) is at most $U^{\varepsilon_{\log}}$ for sufficiently large U . For $U = Z^d$, $d \geq d_{\min}$, set $H = Z^{2\theta_\eta}$ and require

$$0 < \theta_\eta \leq \frac{d_{\min}\varepsilon_{\text{cnt},\text{ht}}}{4(1 + A_{\text{cnt},\eta})}. \quad (11)$$

Then $(1 + H)^{A_{\text{cnt},\eta}} \ll_\eta U^{\varepsilon_{\text{cnt},\text{ht}}}$, so $\#\mathcal{B} \ll_\eta U^{R_* + \varepsilon_{\text{cnt}}}$. The same allocations cover the no-slot count, after taking the finite maximum of its height orders, and the bottom bin has the elementary bound $\#\mathcal{B} \ll_\eta U$. The actual choice of θ_η is deferred to Section 7.

The analytic estimate that will consume this count is proved in Proposition 5.2. After moving the full high-side correction and only then subdividing its retained values, it gives the joint estimate required by (10.13) of [1]. The resulting general exponent of (10.15), relative to $C(7/8)$, is

$$\begin{aligned} E_\tau(d) = a - \frac{7}{8} + h \left(\frac{17}{50} - \frac{1}{6} \right) - al_y - \frac{\ell}{2} + q\ell \\ + d \left(R + \frac{\delta}{2} - \frac{17}{50} \right). \end{aligned} \quad (12)$$

Here R is the ideal count exponent of physical rows in the chosen box. The additional ε_{cnt} , already including count-height absorption, contributes $d\varepsilon_{\text{cnt}}$ to the final Z exponent; it is not included a second time in the joint analytic height factor. For the same a, q, R, d , its change from the old geometry is exactly

$$E_\tau(d) - E_0(d) = \tau \left(\frac{1}{100} + \frac{\delta}{4} + q \right). \quad (13)$$

The explicit margin below is what makes this small increase affordable.

Lemma 4.2. *For the adaptive witness count $R = R_*$, the endpoint $d = h$ satisfies*

$$E_\tau(h) - \left(\beta_* - \frac{7}{8} \right) \leq -\frac{49}{440640} + \frac{15\tau}{8} < 0.$$

Proof. The complete square certificate (20.9) of [1] applies for $0 \leq \delta \leq \alpha$ and $0 \leq x \leq 1/2$. To display its positive margin, set $y = 1/2 - x$, $v = 51 + 41y$, and let E_* denote the old endpoint exponent. Its identity is

$$\begin{aligned} 10368\nu J(-E_*) &= (3 + 5y) \{ (4v\delta - 79)^2 + 49 \} \\ &\quad + 4y \{ 4v\delta[(1 + 3y)(15 + 32y)\delta + 9 - 13y] + 265 + 3485y \}. \end{aligned}$$

Every term is nonnegative, since $9 - 13y \geq 5/2$. Using $v \leq 17(3 + 5y)$ and $J \leq 5/2$ gives

$$E_0(13/16) \leq -\frac{49}{440640}.$$

Increase d from $13/16$ to $h = 13/16 + 3\tau/2$, and then use (13). This gives

$$E_\tau(h) \leq -\frac{49}{440640} + \tau \left(\frac{3}{2}R_* + \delta + q - \frac{1}{2} \right).$$

Since $J \geq \delta P_x$, we have $R_* \leq 17/12 - 3\delta/2$. Together with $q \leq \delta/2$ this bounds the parentheses by $13/8$. Finally, $\beta_* - 7/8 > -\tau/4$, so comparison with $C(\beta_*)$ costs at most a further $\tau/4$. $\square$

The other central ranges have larger fixed margins. The bottom box uses $\delta_0 = 1/50$ and $R = 1$, with old endpoint bound $-7/1200$; the geometric change and comparison cost at most $32\tau/25$. For $d \leq 1/2$ no selected primes are used: take $R = 76/75 - 2\delta/3$. Equation (20.11) gives $-49/14400$, and the change and comparison cost at most $329\tau/400$. Thus put

$$m_A = \frac{49}{440640} - \frac{15\tau}{8} > 10^{-4}. \quad (14)$$

In the adaptive branch the slope in d is positive and less than 2, so its endpoint controls $d \leq h$. Choose

$$0 < \zeta < \min\{1/96, m_A/32\}.$$

Extending to $h + \zeta$ costs less than 2ζ . The same extension still leaves enough compensation primes: since $5\ell - h = 1/48 + 7\tau/2$, on $1/2 \leq d \leq h + \zeta$ we have

$$\frac{\ell}{d} > \frac{1}{5} > \frac{7}{37}.$$

Since $1/5 - 7/37 = 2/185$, this implies the prime-supply condition of Lemma 4.1 with $c_{\text{sup}} = 1/185$ on the entire range where its selected slots are needed. The remaining fixed half of the gap keeps every chosen decrement and rounding allowance strictly inside the supply.

5 Full contours and the joint central estimate

The original statements of Lemmas 10.3–10.6 in [1] restrict their boundary parameter to at least $7/8$. We cannot simply insert a smaller number into those statements. We instead use the exact local replacements and the full representation (7) to establish the affected range checks and contour arguments. These also explain why the count just proved can be applied to retained values without asking an amplitude piece to be holomorphic.

Write $x_r = \text{Re } s$, $w_r = \text{Re } w$, and $z_r = \text{Re } z$. The amplitude ratio $x = q/\delta$ of the previous section remains a separate real parameter.

Proposition 5.1. *The complete correction (6) has the analyticity and full-height bounds required by Definition 10.1 of [1] in its first Euler region and in the region $\mathcal{D}_{2,\tau}$ below. One final excluded set S can make the principal slice satisfy*

$$|H_\eta(s) - 1| \leq \frac{1}{2} \quad (\text{Re } s > \sigma_\tau).$$

The central contours can still be moved for the full row sum, with an external tail whose scale order is fixed independently of the tail order chosen at the end.

Proof. The first Euler region is unchanged:

$$\mathcal{D}_1(\varepsilon_0) : \quad x_r \geq \frac{51}{100}, \quad z_r \geq \frac{17}{50}, \quad w_r \geq -\frac{1}{100}, \quad x_r + w_r \geq 1 + \varepsilon_0.$$

For nonprincipal rows, move first on the global line $x_r = \beta_* + 20e$, and then, only within the buffered heights, to

$$x_r = a + 16e, \quad w_r = 1 - a - 6e,$$

where $e > 0$ is sufficiently small. Throughout the paths, $x_r + w_r \geq 1 + 10e$. The original denominator for a dynamic box is zero-free in its buffered rectangle; global tails remain at $x_r > \beta_*$ and use the global reciprocal bound. These paths need only $\beta_* > 51/100$, rather than $\beta_* \geq 7/8$. The integral and horizontal-edge tail estimates of Lemma 10.2, and the single cumulative height allocation in (10.14), are unchanged. Pointwise amplitude subdivision is introduced only for upper bounds after the complete correction has been moved.

For the principal row enlarge the second Euler region to

$$\mathcal{D}_{2,\tau} : \quad x_r \geq \sigma_\tau, \quad z_r \geq \frac{33}{200}, \quad w_r \geq \frac{19}{20}.$$

This replaces the second region of Definition 10.1 and the principal-slice contraction half-plane in (10.2). In the exact local expressions (7.11), (7.17), the denominators $1-R$, $1-V$, $1-D$ stay uniformly nonzero: for $Q = N\mathfrak{p} \geq 2$,

$$|D| \leq Q^{-\sigma_\tau}, \quad |V| \leq Q^{-99/100}, \quad |R| \leq Q^{-56/25+3\tau/2}.$$

The largest off-row defect exponent and the largest dividing-prime exponent are, respectively,

$$1 - x_r - w_r - 6z_r \leq -\frac{363}{200} + \frac{\tau}{4}, \quad 1 - x_r - w_r \leq -\frac{33}{40} + \frac{\tau}{4}.$$

For clarity, the ramified boundary terms J_j in (7.17), for $j = 1, \dots, 5$, have exponents bounded above by

$$-\frac{73}{40} + \frac{\tau}{4}, \quad -\frac{9}{8} + \frac{3\tau}{4}, \quad \left(-\frac{63}{40} + \frac{3\tau}{4}, -\frac{3}{2} + \tau\right), \quad -\frac{39}{20} + \tau, \quad -\frac{9}{4} + \frac{3\tau}{2}.$$

These and the remaining off-row terms in (7.11), (7.17) are strictly smaller than the corresponding largest exponents above. Writing $\gamma_0 = 163/200 - \tau/4 > 0$ and $\gamma_1 = 33/40 - \tau/4 > 0$, the positive local bounds are $|H_{\mathfrak{p}}| \leq 1 + O(Q^{-1-\gamma_0})$ off u and $|H_{\mathfrak{p}}| \leq 1 + O(Q^{-\gamma_1})$ at primes dividing u . The first product converges. For every $\varepsilon > 0$, the second is $O_\varepsilon((Nu)^\varepsilon)$: outside a fixed finite set each factor is at most Q^ε , and the product of the distinct dividing primes is at most Nu . These estimates are uniform in all imaginary parts and unit phases. Thus the unselected Euler product is normally convergent in the enlarged region with the stated full-height bound. For $u = 1$, enlarge the fixed cutoff in S to make the product uniformly close to 1. The positive product-tail majorant is $O(P_0^{-163/200+\tau/4})$, and gives

$$\sup_{\operatorname{Re} s > \sigma_\tau} |H_\eta(s) - 1| \leq \frac{1}{2}.$$

Further enlargement of S uses the same positive tail bound, without relying on cancellation between complex products.

The complete correction also has the full-row and full-height bounds required by Definition 10.1. On the fixed real boxes where (16.2)–(16.6) apply, $|G_{\mathfrak{p}}| \ll Q^B$ for a fixed B , with height order zero. Bound the unselected Euler factors by the same positive-product majorant and count the finite tuples. Then

$$|H_{\eta,u,Z}(s, w, z)| \ll_\varepsilon (Nu)^\varepsilon Z^{B'},$$

where B' is fixed before the final external tail order N .

To identify the principal slice explicitly, put

$$H_\eta(s) = \prod_{\mathfrak{p} \notin S} H_{\eta,1,\mathfrak{p}}(s, 1, 1/6).$$

At $w = 1, z = 1/6$ the exact local factor gives

$$H_{\eta,\mathfrak{p}}(s) - 1 = O((N\mathfrak{p})^{-2} + (N\mathfrak{p})^{-1-\text{Re } s} + (N\mathfrak{p})^{3-6\text{Re } s}).$$

The slice product therefore converges absolutely and uniformly on compact subsets of $\text{Re } s > 2/3$, and is holomorphic there; this contains $\text{Re } s > \sigma_\tau$.

To bound the complete rows before choosing a tail order, first take $Nu \leq Z^{h+\zeta}$. There are $O(Z^{h+\zeta})$ elements in the fixed unit sector. On each fixed real contour box the displayed positive correction bound, Hecke strip growth, and the appropriate reciprocal estimate give a polynomial majorant after this entire row sum, of the form

$$|F_Z(\vec{t})| \leq C_\eta Z^{B_\eta} (1 + |\vec{t}|)^{J_\eta^*}.$$

Here F_Z includes the scale factors and arithmetic terms but not the external smooth kernel. The principal poles are extracted as residues; the bounds for its remaining contours stay away from those poles. For the unbounded range $Nu > Z^{h+\zeta}$, the absolute row majorant on $(2, 2, z_\infty)$ and its summable dyadic series are proved in Proposition 6.2. They give another fixed polynomial majorant. Taking maxima gives finite B_η, J_η^* for the full collection of discarded integrals and joins before N . A buffered reciprocal is used only on its allocated domain; other tails stay on the global or absolute lines and use their global reciprocal bounds. Extending an integration axis never extends a buffered estimate beyond its rectangle.

For these same fixed W_0, W_1 , the external kernel is

$$e^{-(t_s+t_z)^2} |\mathcal{M}(z_r + it_z)| |\widehat{W}_1(w_r + it_w)|.$$

It is rapidly decreasing in all three variables: the map $(t_s, t_z, t_w) \mapsto (t_s + t_z, t_z, t_w)$ is invertible, and logarithmic integration by parts gives every decay order for the two Mellin transforms. For $\mathcal{M}$, its underlying Fourier transform is smooth at zero and Schwartz at infinity; all the real z intervals here are compact subsets of $(0, \infty)$. Thus the boundary terms vanish. Annular support gives the same conclusion for $\widehat{W}_1$. After the w residue the identical argument applies to the two remaining variables. Any external annular separation profile W is likewise fixed; its all-order Fourier estimates follow by integration by parts, as in Lemma 4.5. Pure twists translate Mellin arguments and append a fixed invertible block triangular height map. Internal coefficient measures controlled only in a fixed weighted L^1 norm remain integrated inside F_Z ; no pointwise trace estimate is inferred for such a measure.

For the truncations below, choose a fixed $c > 0$ no larger than any coordinate multiplier in (10.14). It depends only on the fixed height-allocation matrix and is independent of Z, T_1, N . With k external coordinates, take kernel decay order greater than $N + J_\eta^* + k$. Integrating the resulting positive polynomial majorant outside $\max |t_j| \leq cT_1$, or on a slice $t_j = \pm cT_1$, gives, respectively, the integrated and horizontal-trace bounds

$$O_{\eta,N}(Z^{B_\eta} T_1^{-N}).$$

A bounded real join interval only multiplies this bound by its length. This proves the tails used in the contour argument of Lemma 10.2 and Section 10.3 in the enlarged regions. All absolute interchanges use these full-row majorants. Increasing N changes only which higher seminorms of the same external functions are used; it changes neither B_η, J_η^* nor any internal moment order. Only the retained integral is subsequently subdivided pointwise. $\square$

The next estimate joins the row count to the analytic numerator and all error slots at once. This simultaneous treatment matters because the conductor saving belongs to the whole numerator, and cannot be claimed separately by every slot.

Proposition 5.2. *Put $z_0 = 17/50$ and $\delta = 2a - 1$. Consider central nonprincipal rows in one retained box of Section 8 of [1], at real parts*

$$x_r = a + 16e, \quad w_r = 1 - a - 6e, \quad z_r = z_0.$$

Fix an error-slot set I and an amplitude row set at a retained point. Suppose the row count is at most $C_\eta U^{R+\varepsilon_{\text{cnt}}}$ after (10)–(11), and the mean main-slot amplitude exponent is q , assigning exponent zero to error slots. The corresponding piece of the full correction satisfies

$$\sum_u |L_F^S(w, \chi_\bullet(u)) H_{\eta,u,Z}^{(I)}(s, w, z)| \ll_\eta U^{R+\delta/2+\varepsilon_{\text{cnt}}+\varepsilon_c} \times Z^{\ell(z_0-1/2)+q\ell+\varepsilon_c} (1 + T_1)^{A_\eta}. \quad (15)$$

The central analytic loss ε_c excludes ε_{cnt} and can be made arbitrarily small after the slot system is fixed. The finite height order A_η here includes only the remaining analytic factors; the count height has already been absorbed into ε_{cnt} . It does not change with the external tail order. In particular the amplitude term in the general exponent is $g = q\ell$.

Proof. Use the following local decomposition only in the retained dynamic region. Proposition 16.1 in [1] applies for $51/100 \leq a \leq 1$ without fixing total slot length to $1/6$. Its proof gives

$$H_{\mathfrak{p}} - 1 = O(Q^{-1-10e}) \quad (\mathfrak{p} \nmid u), \quad H_{\mathfrak{p}} - 1 = O(Q^{-10e}) \quad (\mathfrak{p} \mid u).$$

For fixed positive e , a sufficiently large final cutoff makes these local factors nonzero. Write the uncompensated Euler product as $\mathcal{H}_{\eta,u} = \prod_{\mathfrak{p} \notin S} H_{\mathfrak{p}}$, and define here

$$\mathfrak{B}_i = \sum_{\mathfrak{p} \in \mathcal{P}_i(Z)} w_i(N\mathfrak{p}/P_i) (N\mathfrak{p})^{z-1} \frac{G_{\mathfrak{p}}}{H_{\mathfrak{p}}},$$

$$Q_i = - \sum_{\mathfrak{p} \in \mathcal{P}_i(Z)} w_i(N\mathfrak{p}/P_i) (N\mathfrak{p})^{z-1} \overline{\chi_{\mathfrak{p}}(u)}, \quad D_i = \mathfrak{B}_i - Q_i.$$

The conjugate in Q_i is the one in Proposition 16.1, and agrees with the negative physical direction used in Section 4. For $\mathfrak{p} \nmid u$, the local cancellation (16.8) gives

$$G_{\mathfrak{p}}/H_{\mathfrak{p}} + \overline{\chi_{\mathfrak{p}}(u)} = O(Q^{-51/100}).$$

Thus D_i is precisely the error slot jointly estimated in (16.9). Disjoint windows give the exact finite expansion

$$H_{\eta,u,Z} = \mathcal{H}_{\eta,u} \sum_{I \subseteq \{1, \dots, K\}} \prod_{i \in I} D_i \prod_{i \notin I} Q_i = \sum_I H_{\eta,u,Z}^{(I)}. \quad (16)$$

This is (16.10). The nonvanishing needed for these quotients has been proved within the dynamic region only. Outside it, analytic continuation always uses the complete factor G_p of (5).

The buffered estimates for the retained box and Lemma 4.10 give the reflected primitive-numerator small-power bound required by Proposition 16.1. Equation (16.9) then estimates the numerator and every error slot simultaneously:

$$\left| L_F^S(w, \chi_\bullet(u)) \prod_{i \in I} D_i \right| \ll U^{\delta/2 + 12e + \lambda_{\text{joint}}} (1 + T_1)^A \times \prod_{i \in I} P_i^{z_0 - 1/2}. \quad (17)$$

Here $\lambda_{\text{joint}} > 0$ is prescribed before the target, and the preliminary small powers in that proof are reduced to fit it. To see all the charges, put $A_* = a - 1/2 + 6e$. The primitive functional equation contributes $q_f^{A_*}$ and a height polynomial, and the remaining deleted Euler factors cost at most $U^{6e + \lambda}$ because $w_r \geq -6e$. For the full set J_0 of distinct strict ramified labels, (16.11) gives once

$$q_f \ll_S \text{Nu} \prod_{p \in J_0} Q^{-(j_p - 1)}.$$

Each such label has $j_p \geq 2$ and $-w_r - A_* = -1/2$. Hence its allocated local factor is at most $Q^{z_0 - 1/2}$, simultaneously for all these labels; coprime errors have the stronger exponent $z_0 - 51/100$. Divisor-many ramified choices and the other primitive small powers are included in λ_{joint} . This explains the $6e + 6e = 12e$ and removes the need for any separate positive prime-length loss in this bound. The labels are distinct because the underlying windows are disjoint. There is no repeated use of a conductor saving or deletion allowance for individual error slots.

For the remaining slots the normalization is explicitly

$$\tilde{Q}_i = P_i^{-1/2} \sum_{p \in \mathcal{P}_i(Z)} \overline{\chi_p(u)} W_i(Np/P_i) (Np/P_i)^{z-1}, \quad Q_i = -P_i^{z-1/2} \tilde{Q}_i.$$

To form the amplitude boxes, write Lemma 19.1's bound as $|\tilde{Q}_i| \ll U^{\epsilon_1} P_i^{\delta/2 + C_{\text{pre}}}$, with C_{pre} fixed on the real ranges. After K is fixed, put $\ell_{\min} = \ell/K$ and $d_{\max} = h + \zeta$, and choose

$$C_{\text{pre}} + \frac{d_{\max} \epsilon_1}{\ell_{\min}} < \frac{\vartheta}{2}.$$

Increasing the threshold absorbs the fixed constant into the other half of the width, giving $|\tilde{Q}_i| \leq P_i^{\delta/2 + \vartheta}$. The capped subdivision of Section 19.1 then has $0 \leq g_i \leq \delta/2$, with $|\tilde{Q}_i| \geq P_i^{g_i}$ when $g_i > 0$. This prime-estimate cost is already included in ϑ . In an amplitude box take $|\tilde{Q}_i| \leq P_i^{g_i + \vartheta}$, using a zero-exponent upper bound for low-amplitude slots. Assign $g_i = 0$ to each error slot. Then

$$\sum_i \ell_i g_i = q\ell, \quad 0 \leq q \leq \delta/2,$$

by Lemma 19.1. Multiply the main-slot bounds into (17), use $\mathcal{H}_{\eta,u} \ll U^\epsilon$, and sum over the rows. For example, if $\mathcal{H}_{\eta,u} \ll U^{\lambda_H}$, it suffices to choose

$$12e + \lambda_{\text{joint}} + \lambda_H \leq \epsilon_c, \quad \ell\vartheta \leq \epsilon_c.$$

This gives (15), with ε_{cnt} paid separately. There are a fixed 2^K error subsets. Witness presentations and dyads have already been included in the raw count's constant and logarithm; only the remaining amplitude subdivisions and physical norm dyads enter the subsequent multiplicity allowance.

Apply this estimate at each retained point of the already moved complete integral. Its outside real exponent is

$$l_x(1/2 - z_0) + a + z_0 - 1 - al_y - dz_0 + (16 - 6l_y)e.$$

Adding (15), subtracting $C(7/8)$, and using $h = 1 - l_x + \ell$ gives (12) as the ideal exponent. The additional real charges, including the retained reciprocal, are listed in Section 7. All auxiliary heights obey the single allocation (10.14). Neither an error piece nor a pointwise row set is shifted on its own, and neither needs to be holomorphic outside the retained region. This establishes the use of (10.13)–(10.16) for the perturbed geometry. $\square$

6 The principal signal and the outer rows

The central estimate controls the many nonprincipal rows. The principal row has a different role: its two scalar residues produce the signal whose continuation detects zeros. We now identify its positive normalization and the separate sources of error. Small and large nonprincipal rows are then bounded on contours suited to their sizes, so that every physical row is covered.

Proposition 6.1. *After fixing even K , the windows, and the final excluded set, the principal-row double residue produces the signal (18). Its normalizing factor $c_S A_T(Z)$ is positive, with inverse bounded by an arbitrarily small power of Z . The other principal-row integrals and the local approximation retain the respective positive savings $l_y/20$, $h/600$, and κ_P .*

Proof. First the positive tuple bound for all selected factors extends to the new geometry. On the principal rectangle the four error powers in (16.8) of [1] are

$$-x_r, \quad -6z_r, \quad 4 - 5x_r - 6z_r, \quad 1 - w_r - 6z_r,$$

all strictly negative. The positive tuple upper bound for the correction is therefore $Z^{\ell z_r + \varepsilon}$ at every height.

At the principal double-residue slice $u = 1, w = 1, z = 1/6$, define

$$B_p(s) = \frac{G_p(s, 1, 1/6)}{H_p(s, 1, 1/6)}.$$

This slice lies in the second Euler region for the principal row, where H_p is nonzero. The quotient is used only for principal normalization; general contour movement still uses the complete replacement (5). The local identities (16.12)–(16.13) give

$$B_p(s) = -1 + O((N_p)^{-\sigma_\tau}),$$

since $5\sigma_\tau - 3 > \sigma_\tau$.

To normalize the prime sums, set

$$S_i(Z) = \sum_{\mathfrak{p} \in \mathcal{P}_i(Z)} W_i(N\mathfrak{p}/P_i)(N\mathfrak{p})^{-5/6}, \quad A_T(Z) = (-1)^K Z^{-\ell/6} \prod_i S_i(Z).$$

The fixed ray-class prime theorem, Lemma 13.1 of [1], gives

$$S_i(Z) \sim \frac{c_i P_i^{1/6}}{|T| \log P_i}, \quad c_i = \int W_i(y) y^{-5/6} dy > 0.$$

Even K makes $A_T(Z) > 0$, and $A_T(Z) \asymp (\log Z)^{-K}$. Once K is fixed, choose

$$\kappa_P = \frac{\sigma_\tau \ell}{2K} > 0.$$

The local approximation to the double residue has error $O(Z^{-\kappa_P})$, uniformly on the entire global s line.

Repeat the principal contour argument of Lemma 10.5 with the extended region proved above. Keep $s_r = \beta_* + e$, first cross $w = 1$, and then, within its residue, cross $z = 1/6$. No other poles are crossed. Put $r_S = \text{Res}_{\nu=1} \zeta_F^S(\nu) > 0$. The product of the two scalar residues is

$$c_S = \widehat{W}_1(1) \mathcal{M}(1/6) r_S^2 / 6 > 0.$$

The positivity of $\mathcal{M}(1/6)$ is the Fourier-pairing result in (7.2), and the factor $1/6$ is the Jacobian in the residue of $\zeta_F^S(6z)$. The unresidued w integral saves $l_y/20$; the remaining z integral saves $h/600$.

Move the main term of the double residue to $s_r = 2$. It yields the height-independent signal

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{\text{Res}=2} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds. \quad (18)$$

The residue-approximation error is estimated on the original global line. Only its main term is moved to the signal contour. $\square$

Proposition 6.2. *After removing the principal row, the small rows $0 < \text{Nu} \leq Z^{1/100}$ and the large rows $\text{Nu} > Z^{h+\zeta}$ both retain a positive exponent saving. Together with the central range they cover all physical rows.*

Proof. For small rows $u \neq 1$ use

$$x_r = \beta_* + e, \quad w_r = 1/2, \quad z_r = 17/50.$$

They lie in $\mathcal{D}_1(1/3)$ because $\beta_* > \sigma_\tau > 5/6$; the fixed margin $1/3$ thus remains valid in the new contradiction range. At selected primes not dividing u , (16.8) gives $G_{\mathfrak{p}} = O(1)$. At dividing primes its boundary exponents are

$$-1/2, \quad 3/2 - 2x_r, \quad (3/2 - 2x_r, 2 - 3x_r), \quad 2 - 3x_r, \quad 3 - 5x_r.$$

All are at most $1/2$, so $G_{\mathfrak{p}} \ll (N\mathfrak{p})^{1/2}$. Factors from dividing labels are absorbed by the divisor bound. The original numerator and row-count bounds have combined power $63/50 < 2$. The geometric exponent is

$$h \left(\frac{17}{50} - \frac{1}{6} \right) - \frac{l_y}{2} = -\frac{79}{800} + \frac{51\tau}{100}.$$

Allowing the conservative row-count cost $2/100$ and comparing with $C(\beta_*)$ still leaves the saving

$$\frac{63}{800} - \frac{19\tau}{25} > 0.$$

For large rows use the absolute lines $(x_r, w_r, z_r) = (2, 2, z_\infty)$, with fixed $z_\infty > 2$. The exact local formula gives $G_p = O_{z_\infty}(1)$ both off and on the row. The positive unselected product is $O_\varepsilon((Nu)^\varepsilon)$. Absolute ideal counting in each annular slot, before any character cancellation, therefore bounds the full selected-tuple sum by

$$(Nu)^\varepsilon \prod_i P_i^{z_\infty} = (Nu)^\varepsilon Z^{\ell z_\infty}.$$

All three scalar Euler factors are absolutely bounded on these lines. For $V = Z^{h+\zeta}$, dyadic lattice counting gives the convergent series

$$\sum_{Nu > V} (Nu)^{-z_\infty + \varepsilon} \ll \sum_{j \geq 0} (2^j V)^{1 - z_\infty + \varepsilon} \ll V^{1 - z_\infty + \varepsilon}, \quad z_\infty > 1 + \varepsilon.$$

Thus this bound sums the entire infinite large-row range and is uniform in all Mellin heights. The fixed smooth kernel is integrable, so absolute convergence also justifies interchanging this series with the integrals. Since $h = 1 - l_x + \ell$, the resulting exponent tends to minus infinity with z_∞ . Its uniform upper bound is

$$\frac{53}{32} - \frac{3\tau}{4} + (h + \zeta)(1 + \varepsilon) - \zeta z_\infty + \varepsilon.$$

Fix ζ first and then a sufficiently large fixed z_∞ . This gives a uniform positive saving for the large rows. $\square$

7 Choosing the parameters and excluding the gap

We have obtained the low bound, the central estimates, the principal residue, and the outer-row bounds for the same physical probe. Their positive margins can now be combined. The order of the choices is essential: the witness mesh must be fixed before the number of prime slots, whereas the principal approximation becomes smaller as that number increases.

First fix the geometry and the family-wide contradiction gap Δ_τ . This gap is not data attached to any individual target. Fix the central margin and ζ . From its choice,

$$m_A - 2\zeta > \frac{15m_A}{16} = 10^{-4} + \frac{5861}{2350080000}.$$

Divide the central loss allowance between moment-count losses and later analytic subdivision losses, requiring their total to be less than

$$\frac{5861}{4700160000}.$$

This preserves $\mu = 10^{-4}$. Within the first allowance, choose ε_{cnt} and all six terms on the left of (10), with its strict inequality. The two height and logarithmic reserves are allocated now, even though their finite degrees are found only after the target. Fix the witness losses, ν_0 , and ω_{slot} as in Lemma 4.1. The mesh in Lemma 18.1 depends only on the bounded real ranges and ε_m , not on the target or the later K . Then choose a sufficiently large fixed even K such that $2\ell/K < \omega_{\text{slot}}$. The available prime

length stays above $1/5$, and Lemma 4.1 controls both capacities. After fixing the window weights the principal approximation saving is fixed at

$$\kappa_P = \frac{\sigma_\tau \ell}{2K} > 0.$$

The moment losses chosen before K need only fit the central budget. They do not have to be reduced below κ_P : they are used in the central row count, while the principal local approximation is a separate error and uses no moment mesh. Only after fixing K do we shrink the dynamic-box widths, amplitude-box widths, and local small powers. The central charges must fit their remaining allowance; the low-side charges must fit suitable portions of $\Delta_\tau/2$; the principal charges must fit suitable portions of $l_y/20$, $h/600$, and κ_P . More explicitly, with ε_{pr} including the principal small powers and its one normalization allowance, require

$$(1+h)e + \varepsilon_{\text{pr}} < l_y/80, \quad e + \varepsilon_{\text{pr}} < h/2400, \quad e + \varepsilon_{\text{pr}} < \kappa_P/4.$$

Also take e below the detector's bounds for the already prescribed witness losses. These are choices after K ; none asks the earlier moment loss or mesh to be smaller than κ_P . Detection boxes can be refined beyond the fixed moment mesh without changing that mesh or K .

For clarity, the central accounting includes the change of base from $U = Z^d$ to Z . With R retaining its ideal meaning, the complete normalized central real charges are

$$(16 - 6l_y)e + d\varepsilon_{\text{cnt}} + (1+d)\varepsilon_c + d\varepsilon_d + \varepsilon_p + \varepsilon_{\text{norm}}. \quad (19)$$

Here ε_d is exactly the retained reciprocal allowance in (10.16) of [1]: the buffered estimate for $1/L_F^S(s, \eta \overline{\chi_\bullet}(u))$ is $O_\eta(U^{\varepsilon_d})$. It is not a row count. The count contributes $d\varepsilon_{\text{cnt}}$; the joint numerator, simultaneous conductor allocation, deleted factors, unselected product, and main-slot amplitude upper bounds contribute $d\varepsilon_c + \varepsilon_c$, as proved in Proposition 5.2. The power ε_p pays only remaining pointwise or norm-dyadic logarithmic multiplicities, after the witness logarithm has been paid in (10). Finally $\varepsilon_{\text{norm}}$ pays $(c_S A_T(Z))^{-1} \ll_\eta Z^{\varepsilon_{\text{norm}}}$ once for this contribution. It is separate from the primitive or deleted Euler factors. The fixed 2^K expansion and finite character labels cost constants.

If instead one writes $R_{\text{inc}} = R + \varepsilon_{\text{cnt}}$ in (12), its dR_{inc} has already paid the count and the separate $d\varepsilon_{\text{cnt}}$ in (19) must be omitted. We use the ideal convention throughout. Likewise the count height already absorbed in ε_{cnt} is not charged again in a retained analytic height power. The principal estimate has its own single normalization allowance inside ε_{pr} ; it does not incur the central normalization charge a second time. Since $d \leq h + \zeta < 1$, it suffices to arrange

$$(16 - 6l_y)e + \varepsilon_{\text{cnt}} + 2\varepsilon_c + \varepsilon_d + \varepsilon_p + \varepsilon_{\text{norm}} < \frac{5861}{4700160000}.$$

The count share was reserved before K and the remaining terms are chosen after K within the other share. All raw and logarithmic losses are thus paid before the final analytic height is chosen.

Having retained every saving, take

$$m = \frac{1}{4} \min\{\mu, l_y/20, h/600, \kappa_P\} > 0.$$

This separates the earlier moment losses from the later local small powers, avoiding a cycle in which one increases K , shrinks κ_P , and then increases K again. All these positive real parameters and finite structures, including the large absolute line, are fixed before the target η .

For each target choose its fixed arithmetic data and enlarge the excluded set to its final value. Complete every such enlargement first, then use that same final S for the physical probe, Euler correction, c_S , A_T , and the two functions

$$J_\eta(Z) = \frac{I_{\eta, \text{mod}}(Z)}{c_S A_T(Z)}, \quad f_\eta(Z) \text{ as in (18)}.$$

The low bound and logarithmic normalization give

$$|J_\eta(Z)| \ll_\eta Z^{C(\sigma_\tau) + \omega}, \quad \omega = \Delta_\tau/2 \in (0, \Delta_\tau). \quad (20)$$

Now fix all internal moment, seminorm, and Sobolev orders for this datum. Lemma 4.1 gives $A_{\text{cnt}, \eta}$; let $A_{\text{ht}, \eta}$ dominate the finitely many literal detector height orders. The remaining retained analytic height degrees give A_η , excluding the already paid count height. When a remaining bound is expressed in $H = T_1^2$, include twice its degree in A_η . The full-row majorants in Propositions 5.1 and 6.2, with normalization and the remaining finite types, give B_η . All these degrees are finite before a trial θ or any external order is chosen. Subject to the height ceilings below, combining the principal and all nonprincipal estimates gives the same $m > 0$ and, for every fixed N ,

$$|J_\eta(Z) - f_\eta(Z)| \ll_{\eta, N} Z^{C(\beta_*) - m} (1 + T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}. \quad (21)$$

The analytic height occurs in this estimate, not in the definitions of J_η or f_η .

Let $\epsilon_{\text{ht}} > 0$ be the minimum of the finitely many detector height allowances fixed with the preliminary witness losses, and put

$$\theta_{0, \eta} = \frac{d_{\min} \epsilon_{\text{ht}}}{20(1 + A_{\text{ht}, \eta})} > 0.$$

For $T_1 = Z^\theta$ with $0 < \theta \leq \min\{d_{\min}/100, \theta_{0, \eta}\}$ and $U \geq Z^{d_{\min}}$, sufficiently large Z gives

$$T_1 \leq U^{1/100}, \quad (1 + T_1)^{A_{\text{ht}, \eta}} \leq 2^{A_{\text{ht}, \eta}} Z^{d_{\min} \epsilon_{\text{ht}}/20} \leq U^{\epsilon_{\text{ht}}/10}.$$

These are the literal height hypotheses of the pointwise dyadic estimates; they also make the detector error $U^{-189/100 + o(1)} T_1^2$ tend to zero. All finitely many retained physical and witness heights are bounded by a fixed multiple of T_1 , hence by $H = T_1^2 = Z^{2\theta}$ after increasing the threshold. The single matrix allocation (10.14) applies to every added Mellin frequency at once, with total at most $T_1/2$ in each buffered argument. No later estimate renews that allowance.

Choose the actual height by

$$0 < \theta_\eta \leq \min \left\{ \frac{d_{\min}}{100}, \theta_{0, \eta}, \frac{d_{\min} \epsilon_{\text{cnt}, \text{ht}}}{4(1 + A_{\text{cnt}, \eta})}, \frac{m}{4(A_\eta + 1)} \right\}, \quad T_1 = Z^{\theta_\eta}.$$

This both absorbs the count height in its preallocated allowance and gives $(1 + T_1)^{A_\eta} \ll_\eta Z^{m/4}$ for the remaining retained factors. Only then choose a fixed integer N_η with

$$B_\eta - N_\eta \theta_\eta < C(\beta_*) - m/2.$$

Auxiliary external orders in the detector and prime estimates may now also be increased so their discarded parts fit strict power gaps. These are higher-order bounds for the same fixed smooth profiles.

Their constants are absorbed by the final threshold; no internal moment is applied to a newly differentiated external profile. Consequently neither A_η , $A_{\text{cnt},\eta}$, $A_{\text{ht},\eta}$, nor B_η is reset by this last choice. Equation (21) becomes

$$|J_\eta - f_\eta| \ll_\eta Z^{C(\beta_*)-m/2}. \quad (22)$$

Throughout, W , W_0 , W_1 , the slot windows, and the functions J_η , f_η remain fixed. The later orders refer only to higher derivative and decay estimates of the fixed external smooth profiles. The outer slot and presentation types are finite; the internal completed rows, Euler products, and Mellin integrals retain the absolute convergence and majorants proved above.

The quantifier order here is the one needed for this paper: m , ω and all real losses are fixed before η ; for each target the finite degrees and constants precede θ_η ; then come N_η and the lower threshold. Proposition 2.1 of [1] requires the positive savings ω and $m/2$ to be independent of the target, but explicitly permits target-dependent constants, excluded sets, and thresholds. Since neither function contains the auxiliary height, it also permits this target-dependent choice of θ_η , precisely as in Lemma 11.1. No interchange of a target-dependent height degree with a universal target quantifier is being made.

We can now apply the common-signal continuation principle, Proposition 2.1 of [1], which allows $\sigma_0 = \sigma_\tau$. The exponent C has slope one; the multiplier H_η is holomorphic and satisfies $|H_\eta - 1| \leq 1/2$ on the new half-plane; and the same J_η satisfies both (20) and (22). These are exactly the low bound and signal comparison consumed by that principle. It excludes $\beta_* > \sigma_\tau$ by choosing a target with an actual zero sufficiently close to the family supremum. Attainment of the supremum is unnecessary. The assumed gap therefore leads to a contradiction, and

$$\beta_* \leq \sigma_\tau.$$

From primitive Hecke characters to Dirichlet characters

The first step of Proposition 11.3 of [1] extends the Hecke conclusion to imprimitive characters. Passing from a primitive character to an induced presentation changes only finitely many factors $1 - \eta(\mathfrak{p})(N\mathfrak{p})^{-s}$, which are nonzero for $\text{Re } s > 0$.

For the Dirichlet transfer it is useful to name two distinct deletion sets. Let χ be a primitive Dirichlet character of conductor q_χ . Let $T_\mathbb{Q}$ be the rational primes dividing $3q_\chi$, and let T_F consist of all prime ideals above them. These sets are used only in this transfer identity. On ideals coprime to $3q_\chi$ define $\eta(\mathfrak{a}) = \chi(N\mathfrak{a})$, and extend by zero elsewhere. If $\alpha \equiv 1 \pmod{q_\chi \mathcal{O}_F}$, then $N\alpha \equiv 1 \pmod{q_\chi}$, so this defines a finite-order ray class character.

Write χ_{-3} for the quadratic Dirichlet character of conductor 3 associated with $F = \mathbb{Q}(\sqrt{-3})$.

For $p \notin T_\mathbb{Q}$, a split prime has $\chi_{-3}(p) = 1$ and two prime ideals of norm p . Its Hecke factor is $(1 - \chi(p)p^{-s})^{-2}$. An inert prime has $\chi_{-3}(p) = -1$ and one prime ideal of norm p^2 , giving

$$(1 - \chi(p)^2 p^{-2s})^{-1} = (1 - \chi(p)p^{-s})^{-1} (1 + \chi(p)p^{-s})^{-1}.$$

Absolute convergence of the Euler products, followed by meromorphic continuation, yields

$$L_F^{T_F}(s, \eta) = L^{T_\mathbb{Q}}(s, \chi) L^{T_\mathbb{Q}}(s, \chi \chi_{-3}). \quad (23)$$

All deleted factors are nonzero for $\operatorname{Re} s > 0$. In $0 < \operatorname{Re} s < 1$, and at $s = 1 + it$ with $t \neq 0$, both Dirichlet factors have no poles, so a zero of either produces a Hecke zero. For $\operatorname{Re} s > 1$ absolute Euler convergence gives nonvanishing.

At $s = 1$, a possible pole cancellation can occur only when one primitive factor is principal. The other primitive factor is then χ_{-3} , for which

$$L(1, \chi_{-3}) = \int_0^1 \frac{dx}{1+x+x^2} = \frac{\pi}{3\sqrt{3}} > 0.$$

The additional finite Euler factors are also nonzero at $s = 1$. Thus a zero is not hidden by a pole there. The same finite-factor argument handles imprimitive Dirichlet characters. This proves Theorem 1.1, with the strict half-plane $\operatorname{Re} s > \sigma_\tau$.

Remark 7.1 (Further improvements). The explicit central margin invites improvements to the techniques underlying the perturbation. A substantially larger change would require estimates that support the altered lengths: for example, stronger reflected-row energy or a sharper joint numerator and error-slot bound, or greater usable capacity in the inverse and ordinary witness moments. The Euler-region checks, outer-row bounds, and principal approximation would also have to remain compatible with that geometry. These estimates have not been established here for such a change. The present fixed perturbation provides no automatic iteration to the Riemann Hypothesis and no guarantee that successive improvements would reach the critical line. Broader arithmetic questions, including Hilbert’s twelfth problem, are longer-term aspirations for the program; this argument claims no progress on that problem.

AI Closure

This work grew through human–AI dialogue. Human questions and hints guided the choice of a small parameter perturbation and the demand that its consequences be checked for the same finite probe. AI assistance supported the development and explanation of the argument, while the AI for Math plugin supported code generation for computation and the Lean formalization. The mathematical content of the paper is the deduction above from the stated literature inputs; the collaboration does not enlarge those inputs or the theorem’s scope.

The three public terminal statements for the Riemann zeta function, Dirichlet characters, and the stated Hecke family have passed Lean 4.34.1 compilation, clean kernel replay, and standard-axiom checks. The same three statements have also passed the native Comparator core check, including agreement with the declared target statements and their dependencies, permitted-axiom checking, and builtin Lean kernel replay. This recorded Comparator result concerns the native core; the separate Linux sandbox workflow remains in progress. Independent external reproduction and public review remain open.

References

- [1] OpenAI. *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane*. Res > 7/8. September 30, 2026, 199 pp. Official PDF, at Git commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The main inputs here are Sections 2 and 10–20, especially printed pp. 97–107 and 183–196, together with the coefficient, reflection, and zero-detection results of Sections 4–8.
- [2] Bernhard Riemann. *Ueber die Anzahl der Primzahlen unter einer gegebenen Groesse*. Monatsberichte der Berliner Akademie (1859), 671–680. English translation by David R. Wilkins, hosted by the Clay Mathematics Institute. The historical discussion above uses pp. 2–3 for the definition, continuation, and functional equation, and pp. 3–4 for the hypothesis.